

Modern Optimization problems and first-order methods

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Short Bibliography

- X17 - MVA 2021-2022
- PhD in Optimization at University of Twente with Alexandre d'Aspremont, Cristobal Guzman, Sophie Langer and Wouter Koolen
- Post doctoral researcher at ANITI Toulouse with Francois Bachoc, Jerome Bolte and Jean Michel Loubes

What is modern optimization ?

$$\min_{x \in \mathcal{X}} F(x)$$

- F objective function
- $\mathcal{X}$ space of parameters

Examples :

- Neural Network for Image classification $\mathcal{X} = \mathbb{R}^d$
- Game theory $F(x) = \max_{1 \leq j \leq J} f_j(x)$ with $\mathcal{X} = \Delta_d$
- Scheduling $\mathcal{X} = \{x \in \mathbb{R}^d, Ax \leq b\}$

Why first order methods ?

Because it is

- Faster than Zero order methods
- Cheaper to compute than second order methods
- Efficient in practice
- Only option for high dimension ...

Why first order methods ?

- Stochastic Gradient Descent

$$x_{t+1} = x_t - \eta_t G(x_t, \xi_t)$$

- Mirror descent

$$x_{t+1} = \arg \min_{x \in \mathcal{X}} \{ \eta_t \langle G_t(x_t, \xi_t), x \rangle + D_\psi(x, x_t) \}$$

- Frank-Wolfe

$$s_t = \arg \min_{s \in \mathcal{Q}} \{ \langle \nabla f(x_t), s \rangle \}$$

$$x_{t+1} = \left(1 - \frac{2}{t+2} \right) x_t + \frac{2}{t+2} s_t$$

What does (my) current research focus on ?

- Keywords: Non-convex, faster computation, more constraints
- More optimal iterations : Acceleration, optimism from Online learning etc ...

$$x_t = x_{t-1} - \eta_t \nabla F(z_{t-1})$$

$$z_t = (1 - \beta_t) z_{t-1} + \beta_t x_t$$

- Fairness Learning, Majority and Minority

$$L(x) = L_1(x) + L_0(x)$$

- *Optimal Algorithms for Stochastic Complementary Composite Minimization*, Alexandre d'Aspremont, Cristóbal Guzmán and Clement Lezane, 2022
- *Accelerated Mirror Descent for Non-Euclidean Star-convex Functions*, Clement Lezane and Sophie Langer, Wouter M Koolen, 2024
- *When majority rules, minority loses: bias amplification of gradient descent*, François Bachoc, Jerome Bolte, Ryan Boustany, Jean-Michel Loubes, 2024

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